

STUDENT PERFORMANCE ANALYSIS

Detailed Data Science & Data Analytics Project Report

Organization: Cognevance Technologies

Project Level: Level 1 – Easy

1. Project Overview

The Student Performance Analysis project analyzes academic and lifestyle-related student data to identify patterns associated with examination performance. The analysis uses Python-based data analytics techniques for data loading, cleaning, exploratory data analysis, visualization, correlation analysis, insight generation, and recommendations. The project follows the Level 1 requirements provided by Cogneva Technologies.

The supplied project brief requires collection of a student dataset, cleaning and preprocessing, analysis of marks and attendance, creation of bar graphs, histograms and pie charts, investigation of attendance versus academic scores, generation of insights and recommendations, and preparation of a short report.

2. Objectives

- Understand the structure and quality of the student-performance dataset.
- Clean the dataset by checking duplicates and handling missing values.
- Analyze exam scores, attendance, study hours, and selected student characteristics.
- Create meaningful visualizations to communicate the findings.
- Measure the relationship between attendance and exam scores using correlation.
- Measure the relationship between study hours and exam scores.
- Generate practical, evidence-based recommendations for improving academic monitoring.

3. Dataset Description

The dataset contains **1,000 student records** and **16 variables**. The variables cover demographic information, study behavior, digital-media usage, attendance, sleep, diet, exercise, parental education, internet quality, mental-health rating, extracurricular participation, and exam score.

Variable	Description
student_id	Unique student identifier
age	Student age
gender	Gender category
study_hours_per_day	Daily study time
social_media_hours	Daily social-media usage
netflix_hours	Daily Netflix usage
part_time_job	Whether the student has a part-time job
attendance_percentage	Attendance percentage
sleep_hours	Daily sleep duration
diet_quality	Diet quality category
exercise_frequency	Exercise frequency
parental_education_level	Parent/guardian education category
internet_quality	Internet quality category
mental_health_rating	Mental-health rating recorded in the dataset
extracurricular_participation	Extracurricular participation indicator

Variable	Description
exam_score	Final exam score

Dataset quality before cleaning: 91 missing values were present in parental_education_level. No duplicate rows were found.

4. Tools and Technologies

Programming: Python

Libraries: Pandas, NumPy, Matplotlib, Seaborn

Development environment: Jupyter Notebook

Output formats: CSV dataset, PNG visualizations, PDF report, and GitHub repository.

5. Data Cleaning and Preprocessing

The dataset was first loaded into a Pandas DataFrame. Its shape, columns, missing values, duplicate records, information about data types, and descriptive statistics were inspected. Duplicate records were checked and removed. The only missing field identified was parental_education_level, with 91 missing observations. These missing categorical values were replaced using the mode of that column, preserving the student records instead of deleting them.

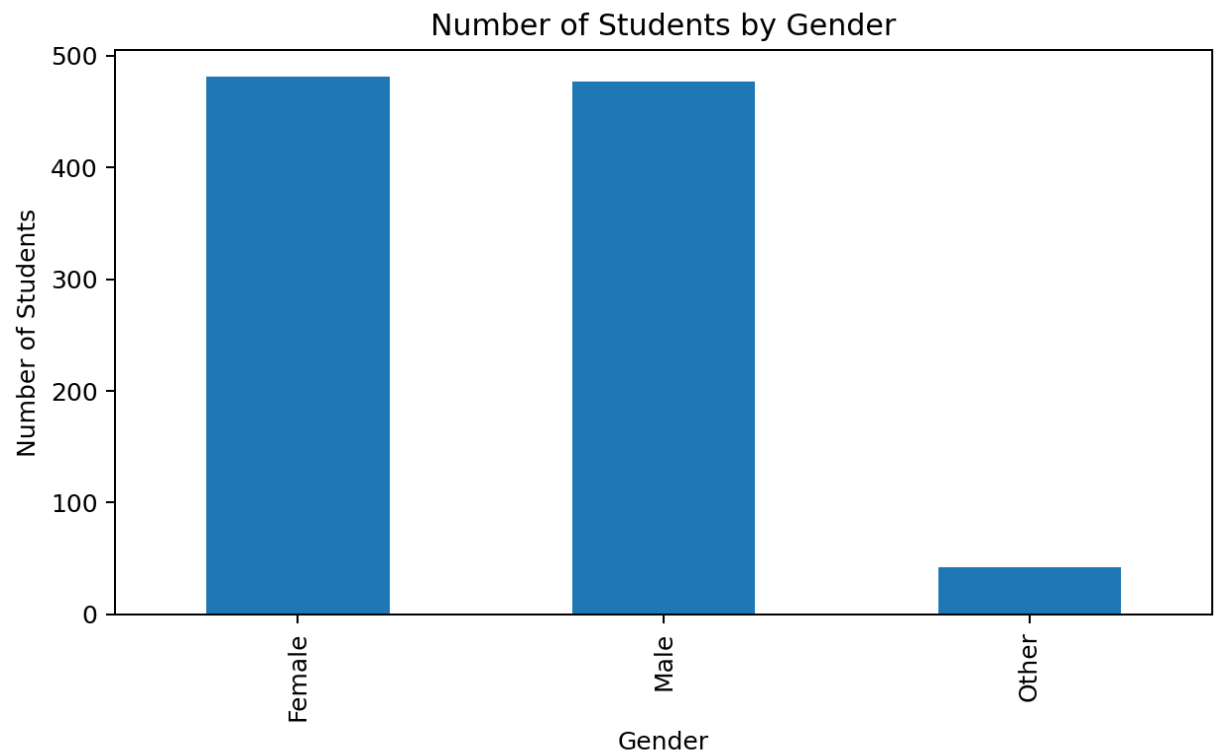
After preprocessing, the analysis dataset contains 1,000 rows, 16 columns, zero missing values, and zero duplicate rows.

6. Exploratory Data Analysis

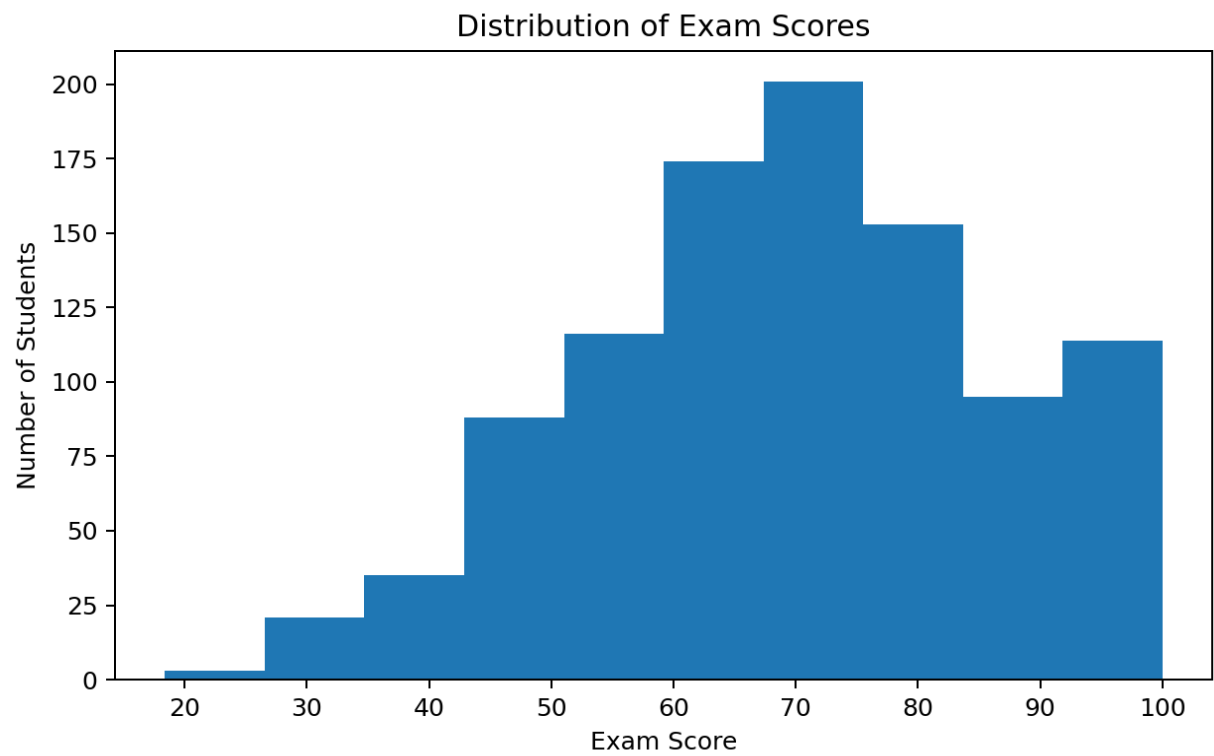
The average exam score is **69.60**, with a minimum of **18.4** and a maximum of **100.0**. Average attendance is **84.13%**, while average study time is **3.55 hours per day**.

Gender distribution consists of 481 Female, 477 Male, and 42 Other students.

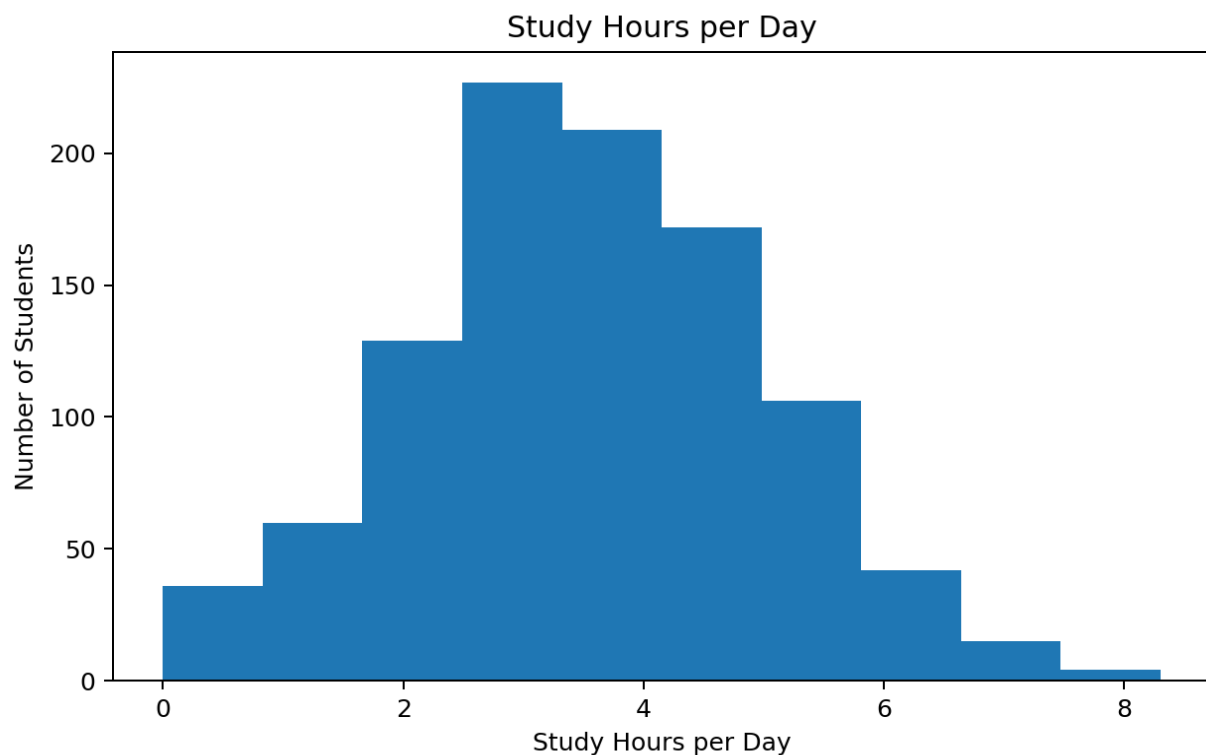
Gender Distribution



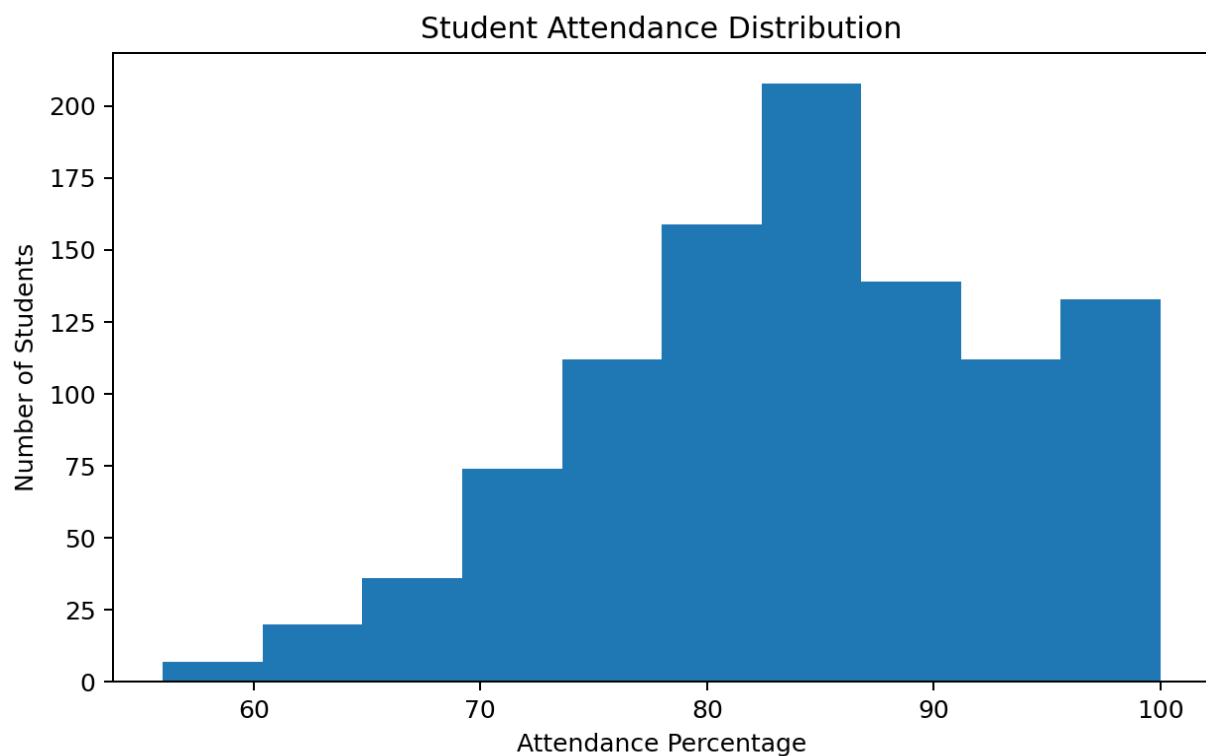
Exam Score Distribution



Study Hours Distribution

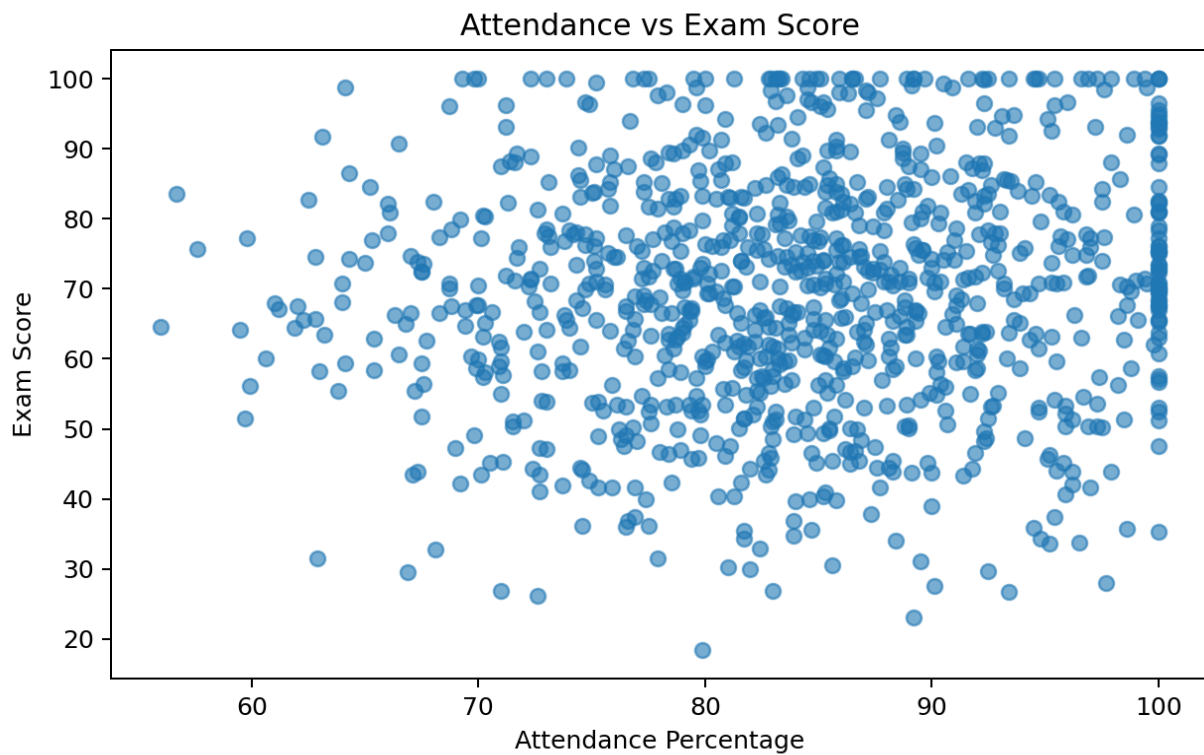


Attendance Distribution



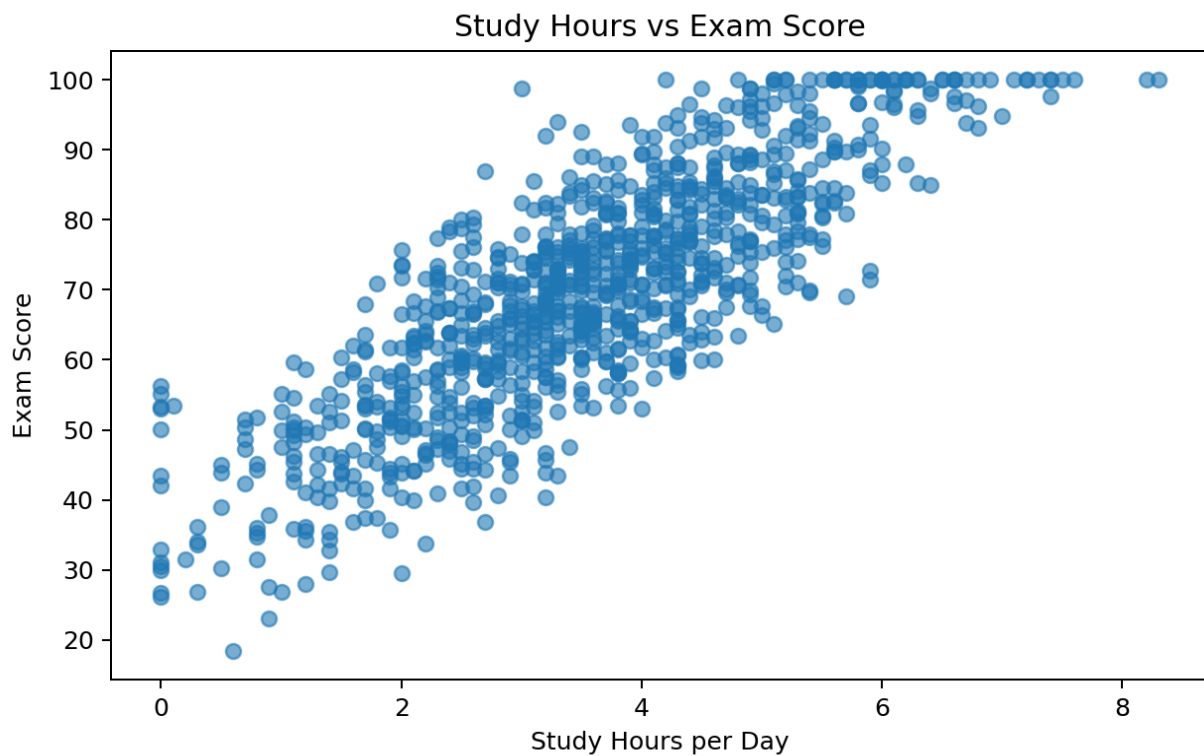
7. Attendance and Academic Performance

The Pearson correlation between attendance percentage and exam score is **0.09**. This is a weak positive linear relationship in this dataset. Therefore, attendance is associated with exam score only weakly in the measured data; it should not be interpreted as proof that attendance alone causes higher marks.



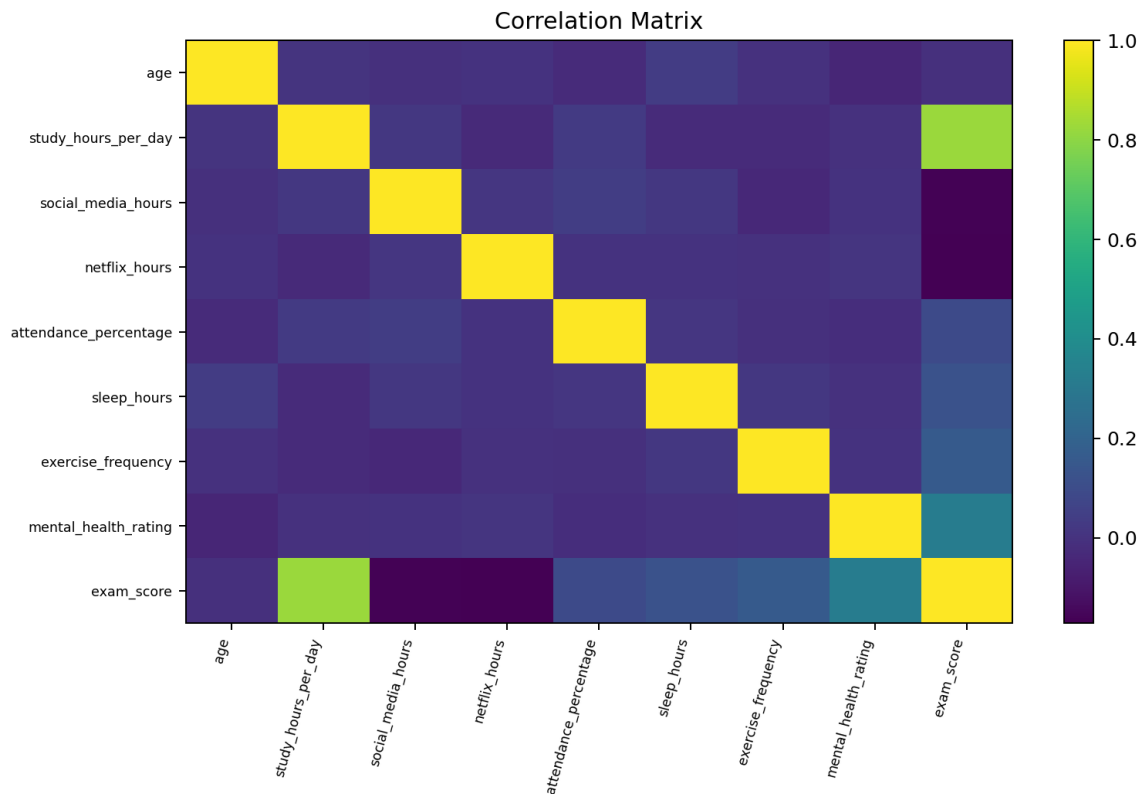
8. Study Hours and Academic Performance

The Pearson correlation between study hours per day and exam score is **0.83**. This is a strong positive linear relationship in the dataset, meaning students with more reported study hours generally tend to have higher exam scores. Correlation does not by itself establish causation.



9. Correlation Analysis

A correlation matrix was generated for the numerical variables. The strongest relationship with exam_score among the variables examined is study_hours_per_day, while attendance_percentage shows a much smaller positive correlation. The matrix is useful for identifying candidate variables for deeper analysis.



10. Key Findings

- The dataset contains 1,000 students and 16 variables.
- The mean exam score is 69.60, with scores ranging from 18.4 to 100.0.
- Average attendance is 84.13%, indicating generally high attendance in the sample.
- Average study time is 3.55 hours per day.
- Attendance and exam score have a correlation of 0.09, indicating a weak positive linear relationship.
- Study hours and exam score have a correlation of 0.83, indicating a strong positive linear relationship.
- 91 missing parental-education values were handled using the column mode, and no duplicate records remained after cleaning.

11. Recommendations

- Encourage consistent study schedules because study hours show the strongest observed relationship with exam score in this dataset.
- Continue monitoring attendance, but avoid using attendance alone as a predictor of academic success because its observed correlation with exam score is weak.
- Identify students with comparatively low exam scores and provide targeted academic support.
- Use periodic data analysis to monitor changes in attendance, study behavior, and academic performance.
- Combine academic indicators with other available variables in future analysis to build a more complete understanding of student performance.

- Use data-driven recommendations carefully: correlations should be treated as associations, not proof of cause and effect.

12. Limitations

- The analysis is based on the supplied dataset and does not establish causal relationships.
- Correlation measures linear association and may not capture nonlinear relationships.
- Only the variables included in the dataset can be evaluated.
- More advanced predictive modeling, statistical testing, and longitudinal data could provide stronger evidence.

13. Conclusion

This project successfully completed the requested student-performance analytics workflow: dataset inspection, cleaning, preprocessing, exploratory analysis, visualization, relationship analysis, insight generation, and recommendations. The analysis found a weak positive association between attendance and exam score (0.09) and a strong positive association between study hours and exam score (0.83). These results suggest that study behavior is an especially important factor to investigate further, while attendance should still be monitored as part of a broader academic-support strategy.

14. Project Deliverables and GitHub Structure

The project brief requires Python/Excel files, visualization charts, the dataset, a project report, and a GitHub repository link. It also requires a README describing the workflow, tools used, and final outcomes.

Recommended repository: `cognevance_StudentPerformanceAnalysis`

Suggested structure:

```
Student_Performance_Analysis/  
■■■■ dataset/  
■ ■■■■ student_habits_performance.csv  
■ ■■■■ student_habits_performance_cleaned.csv  
■■■■ notebook/  
■ ■■■■ student_performance_analysis.ipynb  
■■■■ charts/  
■ ■■■■ gender_distribution.png  
■ ■■■■ exam_score_distribution.png  
■ ■■■■ study_hours_distribution.png  
■ ■■■■ attendance_distribution.png  
■ ■■■■ attendance_vs_exam_score.png  
■ ■■■■ study_hours_vs_exam_score.png  
■ ■■■■ correlation_matrix.png  
■■■■ report/  
■ ■■■■ Student_Performance_Analysis_Detailed_Report.pdf  
■■■■ README.md
```

15. Requirement Checklist

Requirement	Status
Student dataset	Completed
Data cleaning and preprocessing	Completed
Marks/performance analysis	Completed
Attendance analysis	Completed
Bar graph	Completed
Histograms	Completed

Requirement	Status
Pie chart	Not used in the supplied notebook; can be added if r
Attendance vs academic score relationship	Completed
Insights and recommendations	Completed
Short/detailed project report	Completed
Python files/notebook	Completed
Visualization charts	Completed
GitHub repository	To be created/submitted by student

Note: The project brief explicitly lists pie charts among example chart types. The supplied notebook contains a gender bar chart, histograms, scatter plots, and a correlation heatmap, but no pie-chart cell. If your evaluator expects every listed chart type, add a pie chart before submission.

16. Reference to Project Requirements

This report is prepared against the Cognevan Technologies Data Science & Data Analytics project brief. The brief defines Student Performance Analysis as the Level 1 project and specifies the required workflow and deliverables, including dataset collection, cleaning, marks/attendance analysis, charts, attendance-score relationship analysis, insights/recommendations, report, and GitHub submission.